From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

Even untrained random embeddings, when injected into an LLM input, change the model's reasoning behavior in non-trivial ways. Recent *soft prompt* methods optimize trainable virtual-token embeddings to improve mathematical reasoning, but it is unclear whether the gain comes from the *learning* or from the *injection* itself. To separate the two, we remove training entirely: a Random Soft Prompt (RSP) is a continuous vector sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table, and injected as is. Our analysis has two parts. Within one rollout, an exact attention decomposition shows how an RSP can perturb early next-token decisions and then fade as the autoregressive cache grows. Across rollouts, isotropic prompts are direction-agnostic under a variance budget and assign positive probability to every nonempty open set, so independent resampling can reopen early branches; under a separate task-side success assumption, this yields Pass@N growth. The experiments match these claims: zero-training RSP remains competitive with optimized soft prompts, early-token distributions diversify and later stabilize, and Pass@N gains disappear when the same RSP is shared across samples. We close by discussing the implications of this diversity for GRPO training, where trajectory diversity is critical for learning signal quality.

1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. The dominant alternative is parameter-efficient adaptation, in which only a small set of additional parameters is introduced [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* insert learnable continuous embeddings outside the vocabulary into the input, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. The same paradigm has been broadly adopted to enhance mathematical reasoning in recent work [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026], where the common structure is to *optimize* continuous embeddings outside the vocabulary to improve downstream accuracy.

There is, however, a clue that the effect does not come from optimization alone: injecting untrained random continuous embeddings already shifts the output distribution. For example, applying a chat template to a base model that was not fine-tuned for chat causes a format mismatch that degrades reasoning accuracy, but appending random embeddings recovers a substantial portion of that loss. If simple noise were merely perturbing the model, accuracy should worsen instead. This suggests that the effect arises not from *learned content* but from some property of the *injection itself*—a question prior evaluations, focused on end-task accuracy, have not separated from learned content.

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We investigate Random Soft Prompts (RSPs), continuous vectors injected without any training. Each RSP entry is sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table. The role of randomness is not to imitate the embedding distribution; it is to ensure that repeated rollouts receive *independent* perturbations. We answer two questions. First, why can one sampled RSP perturb early next-token decisions and then fade? An exact attention decomposition and two envelope bounds show that the same attention mass both injects and attenuates the perturbation, and that cache growth shrinks the admissible mass late in decoding. Second, why do fresh RSP draws accumulate across trials? Under a fixed variance budget, isotropic perturbations are direction-agnostic, and the Gaussian law additionally assigns positive probability to every nonempty open set. Independent resampling can therefore reopen local top-K entry regions across rollouts, which becomes Pass@N growth once a separate task-side success assumption holds. Finally, we discuss what this diversity implies for GRPO [Shao et al., 2024] training.

Our contributions are as follows.

- We empirically analyze how RSPs affect LLM generation: RSPs raise early-token entropy and stabilize back to baseline at later steps, and combined with temperature sampling produce more diverse reasoning trajectories.
- We explain why untrained RSPs can matter at the transformer level. A single attention-side scalar controls both the size of the injected perturbation and its late-stage decay, and the centered isotropic prompt law neither induces a fixed first-order drift nor excludes any local open branch region.
- We discuss what RSP-induced diversity implies for GRPO training. RSP is a *rank-changing* perturbation while temperature is *rank-preserving*—this difference makes the two methods provide orthogonal exploration gains.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2024] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model's own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2026] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model's reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness guarantees. In the LLM setting, SmoothLLM [Robey et al., 2025] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. **(1)** RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. **(2)** RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. **(3)** RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Random Soft Prompts and Why They Work

This section states the mechanism used in the paper. A *rollout* means one full autoregressive decode under one sampled RSP. Within a rollout, one sampled RSP is fixed, can perturb early next-token decisions through attention, and then fades as the autoregressive cache grows. Across rollouts, independent resampling can accumulate those branch changes into Pass@N gains. We keep only the main statements and intuition here; derivations are deferred to Appendix D.

3.1 Random Soft Prompt

We start with the object being analyzed. Let $\mathbf{W}_E \in \mathbb{R}^{V\times d}$ denote the pretrained token-embedding matrix, where V is the vocabulary size and d is the hidden dimension. Write its entrywise mean and standard deviation as $\mu_E := \frac{1}{Vd}\sum_{i,k}[\mathbf{W}_E]_{i,k}$ and $\sigma_E := \mathrm{std}(\mathbf{W}_E)$. An RSP of length L is a sequence of continuous vectors $\mathbf{H} = [h_1, \ldots, h_L] \in \mathbb{R}^{L\times d}$ drawn independently from

$$h_j \;\sim\; \mathcal{N}(\mu_E \mathbf{1}_d,\; \sigma_E^2 \mathbf{I}_d), \quad j = 1, \ldots, L. \tag{1}$$

The centered form $\bar{h}_j := h_j - \mu_E\mathbf{1}_d \sim \mathcal{N}(0, \sigma_E^2\mathbf{I}_d)$ is a zero-mean isotropic Gaussian. The prompt can be injected as a prefix $[\mathbf{H}; \mathbf{X}]$, infix, or suffix $[\mathbf{X}; \mathbf{H}]$ [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2026]. For repeated-sampling protocols, prompt draws and any decoding randomness across compared trials are assumed independent.

The role of randomness is not to imitate the empirical embedding distribution. It is to make repeated rollouts from the same text prompt receive *independent* perturbations. That independence is what exploration requires.

3.2 Exploration: why the first few tokens move

A single RSP can matter early because it enters the transformer through one scalar that can still be nontrivial before the cache fills. Consider self-attention layer ℓ with n real tokens and L RSP tokens, attention weights $\alpha_{ij}^{(\ell)}$, and value vectors $\mathbf{v}_j^{(\ell)}$ and $\mathbf{v}_{r_j}^{(\ell)}$ on the real and RSP sides. Define the RSP attention mass

$$w_{r,i}^{(\ell)} := \sum_{j'=1}^{L} \alpha_{i,n+j'}^{(\ell)}.$$

The attention output decomposes exactly as

$$\mathbf{x}_i^{(\ell+1)} = (1 - w_{r,i}^{(\ell)})\,\tilde{\mathbf{x}}_i^{(\ell+1)} + \boldsymbol{\eta}_i^{(\ell)}, \quad \tilde{\mathbf{x}}_i^{(\ell+1)} := \sum_{j=1}^{n} \frac{\alpha_{ij}^{(\ell)}}{1-w_{r,i}^{(\ell)}}\mathbf{v}_j^{(\ell)}, \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j'=1}^{L} \alpha_{i,n+j'}^{(\ell)}\mathbf{v}_{r_{j'}}^{(\ell)}. \tag{2}$$

This is an algebraic identity, not an approximation. Because $\|\boldsymbol{\eta}_i^{(\ell)}\| \le w_{r,i}^{(\ell)} \max_{j'} \|\mathbf{v}_{r_{j'}}^{(\ell)}\|$, the same scalar $w_{r,i}^{(\ell)} \in [0,1]$ controls both how much real-token signal is attenuated and how large the random contribution can be.

The only remaining question is whether this scalar can stay meaningfully above zero at early steps. Let $\mathbf{q}_i^{(\ell)}$ be the query and $\mathbf{k}_j^{(\ell)}, \mathbf{k}_{r_{j'}}^{(\ell)}$ the real and RSP keys at layer ℓ.

Proposition 1 (Lower envelope for early exploration)**.** *Define the reverse logit gap* $\Delta_i'^{(\ell)} := \max_{j\in[n]}(\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_j^{(\ell)} - \min_{j'\in[L]}(\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_{r_{j'}}^{(\ell)}$. *Then in scaled dot-product attention,*

$$w_{r,i}^{(\ell)} \;\geq\; \frac{L}{L + n\,\exp(\Delta_i'^{(\ell)}/\sqrt{d})}.$$

The proof is a symmetric softmax bound given in Appendix D. The point of Proposition 1 is modest but useful: if the worst random-token score is not too far below the best real-token score, then early decoding still assigns the random tokens a strictly positive amount of attention mass. That is enough for a single RSP draw to open an alternative early branch within one rollout.

3.3 Annealing: why the effect fades

The same scalar that enables early exploration is also what makes the perturbation fade. Define the forward logit gap

$$\Delta_i^{(\ell)} := \min_{j\in[n]}(\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_j^{(\ell)} - \max_{j'\in[L]}(\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_{r_{j'}}^{(\ell)}, \qquad \Delta_i^{(\ell)} \leq \Delta_i'^{(\ell)}.$$

Theorem 1 (Upper envelope under annealing)**.** *In scaled dot-product attention,*

$$w_{r,i}^{(\ell)} \;\leq\; \frac{L}{L + n\,\exp(\Delta_i^{(\ell)}/\sqrt{d})}.$$

At fixed L and $\Delta_i^{(\ell)}$, the right-hand side is strictly decreasing in n and converges to zero as $n \to \infty$.

Combined with Proposition 1, this gives the two-sided sandwich

$$\frac{L}{L + n\,\exp(\Delta_i'^{(\ell)}/\sqrt{d})} \;\leq\; w_{r,i}^{(\ell)} \;\leq\; \frac{L}{L + n\,\exp(\Delta_i^{(\ell)}/\sqrt{d})}. \tag{3}$$

The theorem does *not* say that the realized mass must decrease step by step: queries, keys, and hidden states co-evolve, so both gaps change over time. What it does say is simpler and more useful: cache growth alone narrows the largest RSP attention mass compatible with a fixed gap. Since Eq. (2) already bounds the random contribution by the same scalar, the perturbation is also bounded by that fixed-gap envelope. Appendix D extends this site-local statement to finite-depth propagation under local Lipschitz control.

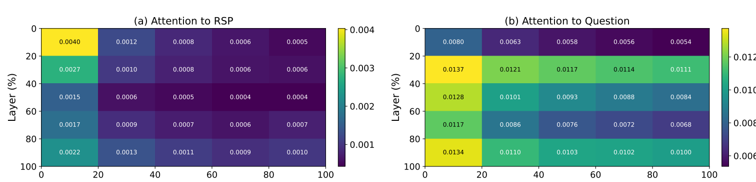

Figure 1: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). (a) RSP-token attention decreases along the reasoning-position axis, consistent with Eq. (3). The additional decrease along the layer axis is a separate empirical observation suggesting that deeper layers sharpen the real-vs-random gap; it does not follow from Theorem 1 alone. (b) Question-token attention remains comparatively uniform. The reported quantity is the per-token mass of Appendix E, equal to $w_{r,i}^{(\ell)}/L$ averaged over heads and samples.

3.4 Performance gain: why independent reruns help

The early and late envelopes explain what one rollout can do. The remaining question is performance: why can *fresh* reruns help more than repeating one fixed perturbation? The answer needs one distributional fact about the prompt law and one separate task-side fact about productive branches.

The first fact is directional. The attention identities above do not require isotropy, but branch opening benefits from a perturbation law that does not under-serve whichever local direction happens to matter for the current prompt.

Theorem 2 (Direction-agnostic minimax coverage)**.** *Let D be any zero-mean distribution on $\mathbb{R}^d$ with covariance Σ_D and trace budget $\operatorname{tr}(\Sigma_D) \leq \rho^2$. Then*

$$\min_{\|b\|=1} \operatorname{Var}_{h\sim D}(b^\top h) = \lambda_{\min}(\Sigma_D) \leq \rho^2/d,$$

with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.

Equality does not require Gaussianity; isotropic covariance is the key property. We choose a Gaussian because it adds the second property we need: positive probability on every nonempty open set. The prompt law therefore does not privilege one direction in second moment, yet it still leaves every local branch-opening region reachable (Proposition 5).

We now formalize branch opening on a first-order surrogate. Let

$$z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}, \qquad a_i := \nabla_{\bar{h}} z_i(0),$$

be the affine approximation of the next-token logits around $\bar{h} = 0$. For a baseline-excluded token i, define the top-K entry region

$$\mathcal{G}_{i,K} \ := \ \left\{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K-1\right\}.$$

Proposition 2 (Conditional top-K entry)**.** *If $\mathcal{G}_{i,K}$ contains a nonempty open set, then the isotropic Gaussian prompt law assigns it positive probability, so*

$$\mu_i := \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0.$$

The mechanism described by Proposition 2 is deliberately limited: it says that a branch can open, not that the opened branch is useful. To connect single-draw branch opening to task accuracy, we add two task-side assumptions. For a problem x, let (M1) each independent RSP run reach a correct trajectory with marginal probability at least $p_{\min}(x) > 0$, and let (M2) the N runs be mutually independent.

Proposition 3 (Multi-path Pass@N)**.** *Under (M1)–(M2), with N independent RSP runs,*

$$\Pr\big(\text{Pass@}N \textit{ on } x\big) \ \geq \ 1 - (1 - p_{\min}(x))^N \ \geq \ 1 - \exp(-N\, p_{\min}(x)).$$

This division of labor is the main interpretive point. The mechanism is responsible only for *admitting* baseline-excluded candidates into the local candidate set. Whether an admitted branch is *productive* is task-dependent, which is why the assumption $p_{\min}(x) > 0$ is separate. It is also why sharing one RSP across all samples removes the accumulation effect: without independent resampling, the same branch is revisited rather than diversified. Section 4 tests exactly this signature.

4 Empirical Validation

We now examine how the explanation in §3 appears in actual LLM generation through three scenes. First, we ask whether RSP preserves baseline accuracy without breaking it, and recovers it in some settings (§4.2). Second, we check whether the early diversification followed by late stabilization—predicted by the two-sided envelope—is observed empirically (§4.3). Finally, we contrast Pass@N between sharing one RSP across samples versus drawing a fresh RSP per sample, directly probing whether *independence* is the key (§4.4).

4.1 Experimental Setup

We evaluate on three mathematical reasoning benchmarks, MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates—seven configurations in total. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity, and the answer-extraction pipeline uses fallbacks to score only the final answer. RSPs follow the definition of §3.1 and are concatenated at one of three positions (prefix, infix, suffix) with a freshly drawn $\mathbf{H}$ for each batch. Full experimental details and per-position results are in Appendix A.

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix). Full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

	MATH-500		**GSM8K**		**AIME24**	
Model	Baseline	RSP	Baseline	RSP	Baseline	RSP
Instruct Models						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
Base Models (Raw Text)						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
Base Models + ChatML (Format Mismatch)						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	**Benchmark**	**Baseline**	**RSP**	**TTSV**	**SoftCoT**	**LTPO**
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	**84.20**	83.80	82.80	83.00
	GSM8K	95.45	**95.68**	95.30	95.00	94.84
	AIME24	13.33	16.67	**23.30**	16.67	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	75.20	75.20	**76.00**	72.40
	GSM8K	85.29	**85.75**	85.70	84.46	84.53
	AIME24	10.00	**20.00**	6.70	6.67	10.00

4.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We first examine how injecting random embeddings affects the model's reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to $+29$ pp. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with the same unified extraction pipeline (Appendix A.2), enabling a format-agnostic comparison despite each method's distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

4.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model's output distribution on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 2 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV). Full statistics are in Appendix A.4.

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple

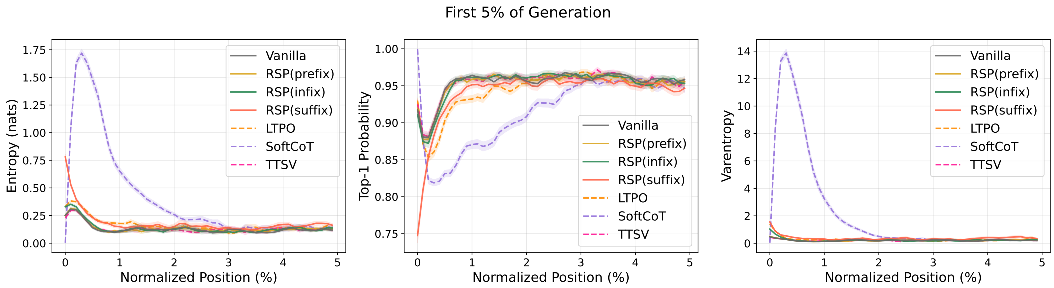

Figure 2: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

reasoning paths can coexist. The change is localized to the initial reasoning stage. In the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

4.4 Does RSP Induce Trajectory Diversity?

Section 4.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP's first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature. We test whether RSP modifies the first-token distribution in a qualitatively different way from temperature scaling, which preserves token rankings while only flattening probabilities. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at τ=1.0 as reference, we compare RSP at τ=1.0 against the baseline at τ=2.0 (the high-temperature ceiling of what flattening alone can achieve) on three complementary metrics: Spearman rank correlation ρ to detect rank reorganization, the probability mass placed outside the baseline's top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change (formal definitions in Appendix B).

Table 3: First-token distribution metrics measured against the baseline at τ=1.0, comparing the baseline at τ=2.0 with RSP at τ=1.0.

Metric	τ=2.0	**RSP**
Spearman ρ	1.0	**0.651**
Mass outside top-10	5.15%	**27.64%**
JS divergence	0.086	**0.231**

Table 3 reports the contrast. Even at τ=2.0, only 5.15% of probability mass falls outside the baseline's top-10. RSP at τ=1.0 reaches 27.64%. The Spearman rank correlation tells the story: because temperature scaling $p_i^{(\tau)} \propto p_i^{1/\tau}$ is a monotone transform that preserves token ranking ($p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$) for any $\tau > 0$, ρ relative to the baseline is exactly 1. RSP, by contrast, reranks the token distribution itself, reducing ρ to 0.651—an effect temperature cannot produce at any τ. The Jensen–Shannon divergence (0.231) quantifies the overall magnitude of this distributional change. Since $\rho < 1$ rules out rank-preserving flattening as the sole cause, the high divergence reflects a structural reshape of the distribution rather than a uniform flattening of the same ranking.

Trajectory-level: semantic diversity. Token-level reranking raises a second question: do first-token perturbations produce

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	**Baseline**	**RSP**
Pairwise cos. dist.	0.0612	**0.0879**
Inter-cluster dist.	0.0183	**0.0982**
Intra-cluster dist.	0.0526	0.0528

semantically diverse reasoning trajectories, or do independently perturbed trajectories converge to semantically similar ones? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at temperature $\tau = 1.0$. Each trajectory is encoded into a dense semantic vector by BGE-M3 [Chen et al., 2024], a sentence-level multilingual embedding model, and we compute three complementary metrics over these embeddings: *pairwise cosine distance* between trajectories captures the overall semantic spread of the trajectory set; *inter-cluster distance* between centroids identified by DBSCAN [Ester et al., 1996], a density-based clustering algorithm that groups nearby trajectories without prespecifying the number of clusters, captures the separation between distinct trajectory groups; and *intra-cluster distance* captures semantic coherence within each group.

Table 4 reveals three concurrent patterns. Pairwise cosine distance rises by about $1.4\times$, indicating that the trajectory set as a whole becomes more semantically diverse. Inter-cluster distance jumps by about $5.4\times$, showing that the trajectories split into substantially more separated groups. Intra-cluster distance is essentially unchanged, indicating that semantic coherence within each group remains comparable to the baseline. RSP also yields larger pairwise distance than baseline on 56 of 64 problems (87.5%).

Outcome-level: Pass@n scaling. Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@n [Chen et al., 2021], the probability that at least one of n sampled solutions is correct. We compare four conditions with $n = 16$ samples per problem on MATH-500 and AIME24: *(i) Baseline,* $\tau = 1.0$: temperature sampling only; *(ii) RSP (fixed seed),* $\tau = 1.0$: a single RSP shared across all samples; *(iii) RSP (indep. seed), greedy*: a different RSP per sample without temperature; and *(iv) RSP (indep. seed),* $\tau = 1.0$: a different RSP per sample combined with temperature sampling.

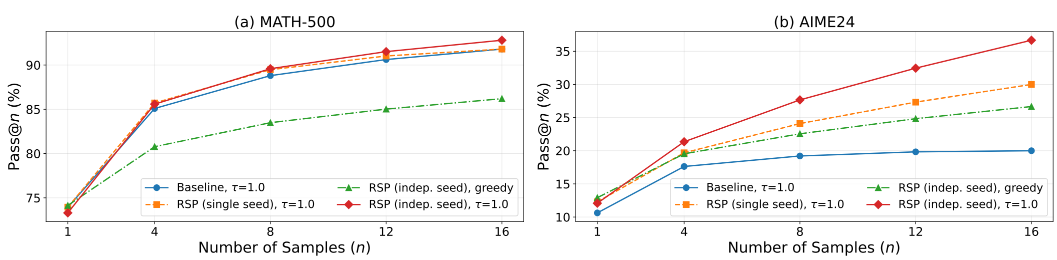

Figure 3: Pass@n scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, $n = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

Condition *(iv)* achieves the highest Pass@n on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $n = 16$. Condition *(ii)* shows no improvement over the baseline, suggesting that the diversity gain likely depends on varying the random embeddings across samples rather than a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and on the more challenging AIME24, *(iv)* achieves higher Pass@1 than the baseline. Together these observations suggest that combining independent RSPs with temperature sampling may yield greater trajectory diversity while preserving single-sample accuracy.

5 Conclusion

We analyzed Random Soft Prompts (RSPs), continuous vectors drawn from an isotropic Gaussian fitted to the entrywise statistics of the pretrained embedding table and injected without any training. The analysis isolates four load-bearing pieces: an exact attention decomposition, a lower envelope for early exploration (§3.2), an upper envelope for late attenuation (§3.3), and a branch-opening argument that becomes Pass@N growth only after adding independent resampling and a separate task-side success assumption (§3.4). The experiments in §4 match these pieces: zero-training RSP remains competitive, early-token distributions diversify and later settle, and the accumulation gain disappears when one RSP is shared across samples. Because RSP is rank-changing while temperature is rank-preserving, the two are complementary, suggesting a sampling-stage remedy for the entropy collapse in GRPO [Shao et al., 2024] beyond DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2026]. Training-time validation against these baselines, extension beyond mathematical reasoning, and a principled choice of RSP length and injection position remain open.

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A Experimental Setup and Full Accuracy Breakdown

Table 5 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to the best-across-positions results summarized in Table 1 in the main text. Table 6 lists the number of RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-Math-1.5B variants).

Table 5: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
Instruct Models				
LLaMA-3.1-8B-Instruct	Baseline	**52.20**	85.75	6.67
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	49.40 (−2.8)	85.97 (+0.2)	**10.00** (+3.3)
	Suffix	49.40 (−2.8)	**86.73** (+1.0)	**10.00** (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	13.33
	Prefix	81.40 (−1.8)	94.92 (−0.5)	13.33 (+0.0)
	Infix	**84.20** (+1.0)	95.60 (+0.2)	**16.67** (+3.3)
	Suffix	83.40 (+0.2)	**95.68** (+0.2)	**16.67** (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	85.29	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	13.33 (+3.3)
	Infix	**75.20** (+2.2)	**85.75** (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	**20.00** (+10.0)
Base Models (Raw Text)				
Qwen2.5-Math-7B	Baseline	**72.20**	84.15	6.67
	Prefix	71.60 (−0.6)	**84.31** (+0.2)	**16.67** (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	**16.67** (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	13.33 (+6.7)
Qwen2.5-Math-1.5B	Baseline	61.40	**77.86**	**16.67**
	Prefix	**64.40** (+3.0)	74.30 (−3.6)	10.00 (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	10.00 (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
Base Models + ChatML (Format Mismatch)				
Qwen2.5-Math-7B	Baseline	52.20	58.30	**23.33**
	Prefix	59.20 (+7.0)	56.41 (−1.9)	3.33 (−20.0)
	Infix	62.00 (+9.8)	69.07 (+10.8)	**23.33** (+0.0)
	Suffix	**70.40** (+18.2)	**75.97** (+17.7)	**23.33** (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	13.33
	Prefix	**63.40** (+29.0)	**67.02** (+29.6)	**20.00** (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	51.00 (+16.6)	50.64 (+13.2)	13.33 (+0.0)

Table 6: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

A.1 RSP Injection Positions and Prompt Templates

Below we show the prompt structure for each injection position across all model types. Blue text indicates RSP embeddings, and purple text indicates special tokens.

A.1.1 Prefix

RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.2 Infix

A description text and special tokens are inserted into the prompt. The special token positions are then replaced with RSP embeddings at the embedding level.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
```

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning.  If these tokens are useful for your case, you can
use them as reference.  If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:
<reserved_special_token_0>...
<reserved_special_token_L>
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning.  If these tokens are useful for your case, you can
use them as reference.  If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:  <special_token_1>...<special_token_L>
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.3 Suffix

For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token. For raw text models, RSP embeddings are concatenated at the end of the prompt.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}[Random Embeddings (L tokens)]<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date:  December 2023
Today Date:  26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}[Random Embeddings (L tokens)]<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
Please reason step by step, and put your final answer within \boxed{}.[Random
Embeddings (L tokens)]
```

A.2 Answer Extraction and Grading

The final answer is extracted from the model output through a fallback chain. Each step is attempted in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. `"final answer is $...$. I hope"` pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following `"the answer is"`
4. Text following `"final answer is"`
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing `$`, `\left`, `\right`, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose`, `rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature $= 0$) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 7 reports the per-benchmark hyperparameters.

Table 7: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 4 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 8 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 8: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle/last-10% segment means with weights $0.1/0.8/0.1$. Extension of Figure 2 (main text) and Figure 4.

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

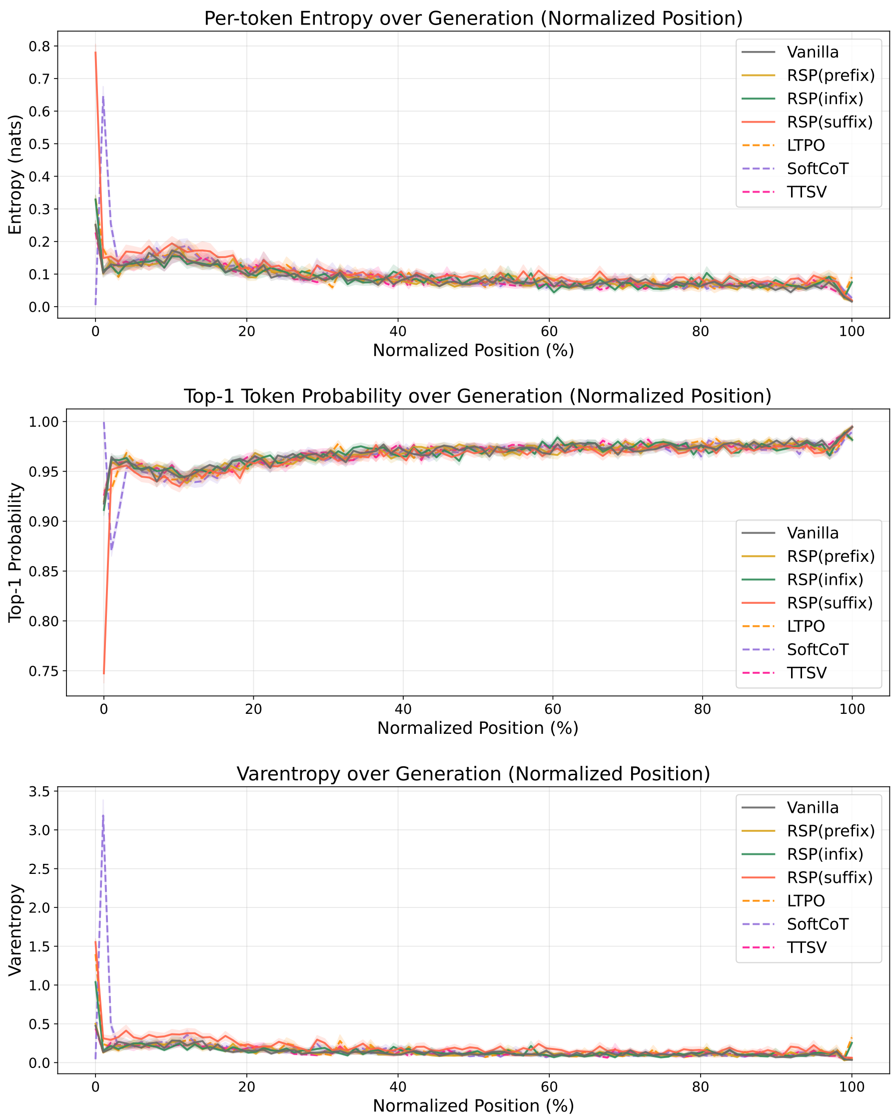


Figure 4: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

B Token-Level Distribution Metrics

This appendix provides formal definitions for the three metrics used in the Token-level analysis of Section 4.4. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the candidate distribution being compared (e.g., the baseline at $\tau = 2.0$ or RSP at $\tau = 1.0$). All metrics are computed analytically from saved logits at the first generation step.

Spearman rank correlation. Spearman's ρ is the Pearson correlation coefficient between the token ranks of two distributions:

$$\rho = \operatorname{Pearson}(\operatorname{rank}(P_{\text{target}}),\ \operatorname{rank}(P_{\text{base}}))\,. \tag{4}$$

Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore indicates rank reorganization beyond what temperature scaling can produce.

Mass outside top-K. Let $\operatorname{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under P_{base}. The probability mass that the candidate distribution places *outside* this set is

$$\operatorname{MassOutside}_K = 1 - \sum_{t \in \operatorname{TopK}(P_{\text{base}})} P_{\text{target}}(t). \tag{5}$$

This directly measures support expansion: how much probability the candidate assigns to tokens that are not in the baseline's preferred set. Reported values use $K = 10$.

Jensen–Shannon divergence. The symmetrized, bounded version of Kullback–Leibler divergence:

$$\operatorname{JS}(P, Q) = \tfrac{1}{2}\operatorname{KL}(P \,\|\, M) + \tfrac{1}{2}\operatorname{KL}(Q \,\|\, M), \qquad M = \tfrac{1}{2}(P + Q). \tag{6}$$

We use base-2 logarithms so that JS $\in [0, 1]$. Tokens absent from the stored top-100 are assigned a sentinel logit (-10^9) before softmax, which is negligible for our setting since the top-5 covers more than 99.9% of the probability mass.

C Prompt-Law Properties Used by the Theory

This appendix isolates the only properties of the RSP law used in the main text: centeredness, direction-agnostic covariance under a variance budget, and positive probability on every nonempty open set. We avoid stronger semantic claims because Section 3.4 needs only these geometric and measure-theoretic facts.

C.1 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let Δ_{ij} be the baseline logit gap between tokens i and j, and write the first-order surrogate as

$$\Delta_{ij}(\bar{h}) = \Delta_{ij} + b_{ij}^{\top}\bar{h}.$$

Proposition 4 (No systematic first-order drift)**.** *If* $\mathbb{E}[\bar{h}] = 0$*, then*

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij}$$

for every token pair (i, j)*. If a deterministic prompt* h_0 *is reused across runs, then*

$$\Delta_{ij}(h_0) - \Delta_{ij} = b_{ij}^{\top}h_0$$

is a fixed directional bias.

Proof. The centered case is immediate from linearity of expectation:

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij} + b_{ij}^{\top}\mathbb{E}[\bar{h}] = \Delta_{ij}.$$

The deterministic case follows by substitution. □

This proposition rules out a run-averaged first-order bias. It does not say that individual prompt draws leave the logits unchanged.

C.2 Proof of Theorem 2

For a unit vector b, the first-order perturbation energy available along b is $\mathrm{Var}_{h\sim D}(b^{\top}h) = b^{\top}\Sigma_D b$. The worst-case directional energy is therefore the minimum Rayleigh quotient of Σ_D.

Proof. For any symmetric covariance matrix Σ_D,

$$\min_{\|b\|=1} b^{\top}\Sigma_D b = \lambda_{\min}(\Sigma_D).$$

Let the eigenvalues of Σ_D be $\lambda_1, \ldots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \le \frac{1}{d}\sum_{m=1}^{d}\lambda_m = \frac{\mathrm{tr}(\Sigma_D)}{d} \le \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. □

The proof uses only covariance. We choose a Gaussian law for RSP because it adds the open-set reachability property proved next.

C.3 Open-Set Reachability of Isotropic Gaussian Prompts

Proposition 5 (Open-set reachability)**.** *Let* $\bar{h} \sim \mathcal{N}(0, \sigma^2\mathbf{I}_d)$ *with* $\sigma > 0$*. Then every nonempty open set* $U \subseteq \mathbb{R}^d$ *satisfies*

$$\Pr(\bar{h} \in U) > 0.$$

Proof. The Gaussian density

$$(2\pi\sigma^2)^{-d/2}\exp\left(-\frac{\|\bar{h}\|^2}{2\sigma^2}\right)$$

is strictly positive at every point of $\mathbb{R}^d$. Every nonempty open set contains a ball of positive Lebesgue measure, and integrating a strictly positive density over that ball gives positive probability. □

This is the only support property used in Proposition 2.

D Proofs for the Transformer-Level Mechanism

This appendix follows the same order as the main theory section: exploration, annealing, and performance gain. The prompt-law facts used before these proofs are collected in Appendix C.

D.1 Exploration: attention decomposition and perturbation size

Proposition 6 (Attention decomposition)**.** *Let $\alpha_{ij}^{(\ell)}$ be the attention weights for query position i at layer ℓ, with n real tokens and L random tokens. Define*

$$w_{r,i}^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)}$$

and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(\ell)} := \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}}.$$

Then

$$\mathbf{x}_i^{(\ell+1)} = (1 - w_{r,i}^{(\ell)})\,\tilde{\mathbf{x}}_i^{(\ell+1)} + \boldsymbol{\eta}_i^{(\ell)},$$

where

$$\tilde{\mathbf{x}}_i^{(\ell+1)} := \sum_{j=1}^{n} \tilde{\alpha}_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} \qquad \textit{and} \qquad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

Proof. The attention output is

$$\mathbf{x}_i^{(\ell+1)} = \sum_{j=1}^{n} \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

By definition, $\sum_{j=1}^{n} \alpha_{ij}^{(\ell)} = 1 - w_{r,i}^{(\ell)}$, so

$$\sum_{j=1}^{n} \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \sum_{j=1}^{n} \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)})\,\tilde{\mathbf{x}}_i^{(\ell+1)}.$$

Substituting this identity gives the decomposition. □

Corollary 1 (Magnitude scales with random-token attention)**.** *For every realization of the random values,*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \le w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

Proof. By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(\ell)}\| = \left\| \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)} \right\| \le \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \|\mathbf{v}_{r_j}^{(\ell)}\| \le w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

□

Corollary 1 is the only quantitative fact needed in the main text. No independence assumption between attention weights and random keys is required. Once $w_{r,i}^{(\ell)}$ is small, the random contribution must be small as well.

D.2 Annealing: upper and lower envelopes

Proof of Theorem 1 (upper envelope). Write the scaled scores as $s_j^{(\ell)} := (\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_j^{(\ell)}/\sqrt{d}$ for $j \in [n]$ and $s_{r_{j'}}^{(\ell)} := (\mathbf{q}_i^{(\ell)})^\top \mathbf{k}_{r_{j'}}^{(\ell)}/\sqrt{d}$ for $j' \in [L]$, and let $s_{r,\max}^{(\ell)} := \max_{j'} s_{r_{j'}}^{(\ell)}$ and $R^{(\ell)} := \sum_{j'=1}^{L} \exp(s_{r_{j'}}^{(\ell)})$. The forward-gap definition gives, for every $j \in [n]$,

$$s_j^{(\ell)} \;\geq\; s_{r,\max}^{(\ell)} + \Delta_i^{(\ell)}/\sqrt{d}.$$

Since $\exp(s_{r,\max}^{(\ell)}) \geq R^{(\ell)}/L$ (max dominates average), summing over the n real keys yields

$$\sum_{j=1}^{n} \exp(s_j^{(\ell)}) \;\geq\; n\,\exp(\Delta_i^{(\ell)}/\sqrt{d})\,\exp(s_{r,\max}^{(\ell)}) \;\geq\; \frac{n\,\exp(\Delta_i^{(\ell)}/\sqrt{d})}{L}\,R^{(\ell)}.$$

Plugging this into the softmax-normalized RSP mass,

$$w_{r,i}^{(\ell)} = \frac{R^{(\ell)}}{\sum_{j=1}^{n}\exp(s_j^{(\ell)}) + R^{(\ell)}} \;\leq\; \frac{R^{(\ell)}}{(n\,\exp(\Delta_i^{(\ell)}/\sqrt{d})/L)\,R^{(\ell)} + R^{(\ell)}} = \frac{L}{L + n\,\exp(\Delta_i^{(\ell)}/\sqrt{d})}.$$

Proof of Proposition 1 (lower envelope). The argument is symmetric. Let $s_{r,\min}^{(\ell)} := \min_{j'} s_{r_{j'}}^{(\ell)}$. The reverse-gap definition gives, for every $j \in [n]$,

$$s_j^{(\ell)} \;\leq\; s_{r,\min}^{(\ell)} + \Delta_i'^{(\ell)}/\sqrt{d}.$$

Since $\exp(s_{r,\min}^{(\ell)}) \leq R^{(\ell)}/L$ (min is dominated by average),

$$\sum_{j=1}^{n} \exp(s_j^{(\ell)}) \;\leq\; \frac{n\,\exp(\Delta_i'^{(\ell)}/\sqrt{d})}{L}\,R^{(\ell)},$$

and therefore

$$w_{r,i}^{(\ell)} = \frac{R^{(\ell)}}{\sum_{j=1}^{n}\exp(s_j^{(\ell)}) + R^{(\ell)}} \;\geq\; \frac{L}{L + n\,\exp(\Delta_i'^{(\ell)}/\sqrt{d})}.$$

D.3 Annealing: corollaries on cache growth

The two envelopes yield the corollaries below.

Corollary 2 (Sequence-wise annealing of the fixed-gap envelope)**.** *For fixed L and $\Delta_i^{(\ell)}$, the upper bound*

$$f(n) = \frac{L}{L + n\exp(\Delta_i^{(\ell)}/\sqrt{d})}$$

is strictly decreasing in n. Cache growth alone therefore narrows the largest RSP attention mass compatible with that gap.

During autoregressive decoding $\Delta_i^{(\ell)}$ also moves because queries, keys, and hidden states co-evolve. The accurate reading is not "the realized mass decreases at every step" but "the envelope compatible with a fixed gap shrinks as the cache grows."

Proof. Differentiate $f(n)$ with respect to n:

$$f'(n) = -\frac{L\exp(\Delta_i^{(\ell)}/\sqrt{d})}{\Big(L + n\exp(\Delta_i^{(\ell)}/\sqrt{d})\Big)^2} < 0.$$

Thus $f(n)$ is strictly decreasing. □

Corollary 3 (Vanishing random contribution under annealing)**.** *If $n \to \infty$ while L and $\Delta_i^{(\ell)}$ remain fixed, $w_{r,i}^{(\ell)} \to 0$, and for every realization of the random values*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j\in[L]} \|\mathbf{v}_{r_j}^{(\ell)}\| \to 0.$$

Proof. $w_{r,i}^{(\ell)} \to 0$ by Corollary 2. Apply Corollary 1 to bound $\|\boldsymbol{\eta}_i^{(\ell)}\|$. The maximum norm is finite for any realization of finitely many sampled values. □

D.4 Annealing: finite-depth propagation under local Lipschitz control

A transformer block applies an attention sublayer followed by a feed-forward sublayer with residual connections:

$$\mathbf{x}_i^{(\ell,\text{attn})} = \mathbf{x}_i^{(\ell)} + \text{Attn}^{(\ell)}(\mathbf{x}_i^{(\ell)}), \qquad \mathbf{x}_i^{(\ell+1)} = \mathbf{x}_i^{(\ell,\text{attn})} + F^{(\ell)}(\mathbf{x}_i^{(\ell,\text{attn})}).$$

Let $\mathbf{x}_i^{(\ell,\text{attn,RSP})}$ and $\mathbf{x}_i^{(\ell,\text{attn,base})}$ denote the post-attention states under RSP and the unperturbed forward pass at the same input $\mathbf{x}_i^{(\ell)}$. To avoid clashing with the RSP length L, we write Lipschitz constants of the FFN sublayer as $\kappa^{(\ell)}$ and of the full block as $\kappa_{\text{B}}^{(\ell)}$, and the model depth as Λ.

Lemma 1 (Per-block FFN propagation)**.** *The post-attention difference satisfies*

$$\|\mathbf{x}_i^{(\ell,\textit{attn,RSP})} - \mathbf{x}_i^{(\ell,\textit{attn,base})}\| \le \|\boldsymbol{\eta}_i^{(\ell)}\| + w_{r,i}^{(\ell)}\|\tilde{\mathbf{x}}_i^{(\ell+1)}\|.$$

If, in addition, $F^{(\ell)}$ is $\kappa^{(\ell)}$-Lipschitz on a bounded set containing both $\mathbf{x}_i^{(\ell,\textit{attn,RSP})}$ and $\mathbf{x}_i^{(\ell,\textit{attn,base})}$, then

$$\|\mathbf{x}_i^{(\ell+1,\textit{RSP})} - \mathbf{x}_i^{(\ell+1,\textit{base})}\| \le (1+\kappa^{(\ell)})\big(\|\boldsymbol{\eta}_i^{(\ell)}\| + w_{r,i}^{(\ell)}\|\tilde{\mathbf{x}}_i^{(\ell+1)}\|\big).$$

Proof. The unperturbed attention sublayer produces $\mathbf{x}_i^{(\ell,\text{attn,base})} = \mathbf{x}_i^{(\ell)} + \tilde{\mathbf{x}}_i^{(\ell+1)}$, while the RSP attention sublayer produces $\mathbf{x}_i^{(\ell,\text{attn,RSP})} = \mathbf{x}_i^{(\ell)} + (1-w_{r,i}^{(\ell)})\tilde{\mathbf{x}}_i^{(\ell+1)} + \boldsymbol{\eta}_i^{(\ell)}$ by Proposition 6. Subtracting,

$$\mathbf{x}_i^{(\ell,\text{attn,RSP})} - \mathbf{x}_i^{(\ell,\text{attn,base})} = \boldsymbol{\eta}_i^{(\ell)} - w_{r,i}^{(\ell)}\tilde{\mathbf{x}}_i^{(\ell+1)},$$

and the triangle inequality gives the first bound. For the second, write the post-FFN difference as

$$\big[\mathbf{x}_i^{(\ell,\text{attn,RSP})} + F^{(\ell)}(\mathbf{x}_i^{(\ell,\text{attn,RSP})})\big] - \big[\mathbf{x}_i^{(\ell,\text{attn,base})} + F^{(\ell)}(\mathbf{x}_i^{(\ell,\text{attn,base})})\big].$$

By Lipschitz continuity, $\|F^{(\ell)}(\mathbf{x}^{(\ell,\text{attn,RSP})}) - F^{(\ell)}(\mathbf{x}^{(\ell,\text{attn,base})})\| \le \kappa^{(\ell)}\|\mathbf{x}_i^{(\ell,\text{attn,RSP})} - \mathbf{x}_i^{(\ell,\text{attn,base})}\|$. The triangle inequality combined with the first bound gives the second. □

Corollary 4 (Cross-block propagation control)**.** *Define $\Delta^{(\ell)} := \mathbf{x}_i^{(\ell,\textit{RSP})} - \mathbf{x}_i^{(\ell,\textit{base})}$ with $\Delta^{(0)} = \mathbf{0}$ (matched inputs at layer 0), and let $\kappa_{\text{B}}^{(\ell)}$ denote a common local Lipschitz constant valid for both the baseline block map $B_{\textit{base}}^{(\ell)} : \mathbf{x} \mapsto \mathbf{x} + \text{Attn}^{(\ell)}(\mathbf{x}) + F^{(\ell)}(\mathbf{x} + \text{Attn}^{(\ell)}(\mathbf{x}))$ and the RSP-augmented block map $B_{\textit{RSP}}^{(\ell)}$ (which replaces $\text{Attn}^{(\ell)}$ with the attention computation that includes the random tokens) on a ball containing both trajectories. Such a constant exists for standard transformer blocks because Attn, LayerNorm, and the FFN sublayer are each locally Lipschitz on any bounded set [Xiong et al., 2020, Kim et al., 2021], and the random-key augmentation only adds a finite number of additional softmax inputs at every layer. Then*

$$\|\Delta^{(\ell+1)}\| \le (1+\kappa_{\text{B}}^{(\ell)})\,\|\Delta^{(\ell)}\| + (1+\kappa^{(\ell)})\big(\|\boldsymbol{\eta}_i^{(\ell)}\| + w_{r,i}^{(\ell)}\|\tilde{\mathbf{x}}_i^{(\ell+1)}\|\big).$$

Iterating across Λ blocks with $\Delta^{(0)} = \mathbf{0}$ yields

$$\|\Delta^{(\Lambda)}\| \le \sum_{\ell=0}^{\Lambda-1}\prod_{\ell'=\ell+1}^{\Lambda-1}(1+\kappa_{\text{B}}^{(\ell')}) \cdot (1+\kappa^{(\ell)})\big(\|\boldsymbol{\eta}_i^{(\ell)}\| + w_{r,i}^{(\ell)}\|\tilde{\mathbf{x}}_i^{(\ell+1)}\|\big).$$

Proof. Decompose the cross-block discrepancy at layer $\ell+1$ into two contributions:

$$\Delta^{(\ell+1)} = \underbrace{B_{\text{RSP}}^{(\ell)}(\mathbf{x}^{(\ell,\text{RSP})}) - B_{\text{RSP}}^{(\ell)}(\mathbf{x}^{(\ell,\text{base})})}_{\text{(I) propagation of prior } \Delta^{(\ell)}} + \underbrace{B_{\text{RSP}}^{(\ell)}(\mathbf{x}^{(\ell,\text{base})}) - B_{\text{base}}^{(\ell)}(\mathbf{x}^{(\ell,\text{base})})}_{\text{(II) RSP-induced injection at block } \ell},$$

where $B_{\text{RSP}}^{(\ell)}$ replaces $\text{Attn}^{(\ell)}$ with the RSP-augmented attention. Term (I) is bounded via the block Lipschitz constant: $\|(I)\| \le \kappa_{\text{B}}^{(\ell)}\|\Delta^{(\ell)}\|$. With residuals included one obtains the factor $(1+\kappa_{\text{B}}^{(\ell)})$ when also propagating $\Delta^{(\ell)}$ itself. Term (II) is exactly the per-block injection from Lemma 1, bounded by $(1+\kappa^{(\ell)})\big(\|\boldsymbol{\eta}_i^{(\ell)}\| + w_{r,i}^{(\ell)}\|\tilde{\mathbf{x}}_i^{(\ell+1)}\|\big)$. The triangle inequality gives the recursion, and unrolling with $\Delta^{(0)} = \mathbf{0}$ yields the displayed sum. □

Boundedness of $\kappa^{(\ell)}$ for standard architectures. Modern transformer blocks combine LayerNorm, GELU or SwiGLU activations, and linear projections. LayerNorm has Lipschitz constant on the order of $\|\gamma\|/\inf\|x\|$ on every set bounded away from the origin, where γ is the LayerNorm scale parameter [Xiong et al., 2020]. GELU and SwiGLU are smooth and therefore locally Lipschitz on bounded activation regions, and linear projections have Lipschitz constant equal to the spectral norm of the weight matrix. The composed self-attention sublayer has a Lipschitz constant determined by the spectral norms of $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ and the inverse-temperature β as analyzed by Kim et al. [2021]. On the bounded activation regions traversed during a single forward pass, the per-block Lipschitz constants in Corollary 4 are therefore finite. For a fixed depth Λ this yields a finite cumulative bound on $\|\Delta^{(\Lambda)}\|$; stronger depth-uniform non-expansion would require additional contraction assumptions not imposed here. This is the formal sense in which the injected perturbation stays controlled across a finite forward pass, while the size of each new injection is set by $w_{r,i}^{(\ell)}$.

D.5 Performance gain: from local admission to Pass@N

The main text §3.4 uses a first-order surrogate to state two claims: a baseline-excluded token can enter the local top-K set, and independent reruns can turn such local openings into Pass@N gains. The proofs below use only one support condition on the centered prompt law D:

$$\text{(S1)} \qquad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

Proposition 5 shows that the isotropic Gaussian law of §3.4 satisfies (S1).

D.5.1 Autoregressive Formulation

In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^{T} p(y_t \mid x, y_{<t}),$$

where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single global distribution.

D.5.2 Local Linearization of Logits under RSP

Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector from the RSP definition (Eq. (1) in §3.1), treated as a per-token surrogate. The actual implementation uses a length-L prompt $\mathbf{H} = (h_1, \ldots, h_L)$ with L independent draws; the arguments below isolate how one local perturbation can shift the logits. We model the perturbed logits as a function $z_i(\bar{h})$, assuming local smoothness in a neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_h z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local affine surrogate

$$z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

This approximation is used as a first-order surrogate for branch opening, not as an exact global model of the transformer logits.

D.5.3 Proof of Proposition 2

Let

$$\mathcal{G}_{i,K} \;:=\; \big\{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K-1\big\}.$$

Proof. Assume $\mathcal{G}_{i,K}$ contains a nonempty open set U. By (S1), $D(U) > 0$. Since $U \subseteq \mathcal{G}_{i,K}$,

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) \;\geq\; D(U) \;>\; 0.$$

This is exactly the event that token i enters the surrogate top-K set.

A useful sufficient condition is the existence of some $\bar{h}_0$ at which token i strictly outranks all but at most $K-1$ other tokens in the affine surrogate. By continuity, a neighborhood of $\bar{h}_0$ is then contained in $\mathcal{G}_{i,K}$. $\square$

D.5.4 Proof of Proposition 3

This proposition does not derive $p_{\min}(x) > 0$ from the mechanism. It only converts a task-side lower bound and independence into Pass@N growth.

Proof. Let A_n be the event that the n-th independent RSP run reaches a correct trajectory on problem x, and write $p_n(x) := \Pr(A_n) \geq p_{\min}(x)$ as in (M1). Under (M2),

$$\Pr\Big(\bigcap_{n=1}^{N} A_n^c\Big) = \prod_{n=1}^{N}(1 - p_n(x)) \leq \prod_{n=1}^{N}(1 - p_{\min}(x)) = (1 - p_{\min}(x))^N.$$

Taking complements gives

$$\Pr\big(\text{Pass@}N \text{ on } x\big) = \Pr\Big(\bigcup_{n=1}^{N} A_n\Big) \geq 1 - (1 - p_{\min}(x))^N.$$

The bound $1 - x \leq e^{-x}$ for $x \in [0, 1]$ then yields

$$1 - (1 - p_{\min}(x))^N \;\geq\; 1 - e^{-N p_{\min}(x)}.$$

$\square$

E Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 1 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

E.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer ℓ, head h, from query position i to key position j, satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\mathrm{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^{H} \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \qquad \mathrm{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^{H} \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \tag{7}$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

E.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \mathrm{PTM}_\bullet[\ell, i]. \tag{8}$$

Figure 1 reports the sample average of Eq. (8) over the 500 valid samples.

E.3 Excluding the `\boxed{...}` Span

The reasoning position range terminates strictly before the final `\boxed{...}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{...}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

E.4 Excluding the Final Two Layers

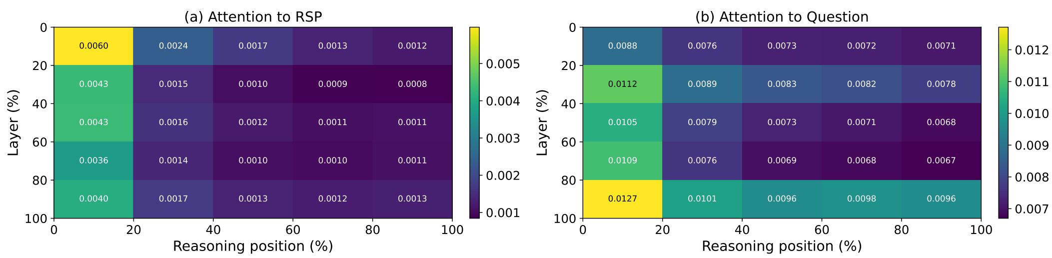
The layer axis further excludes the last two transformer layers. This choice is motivated by a standard late-layer effect rather than by any model-specific tuning decision.

Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

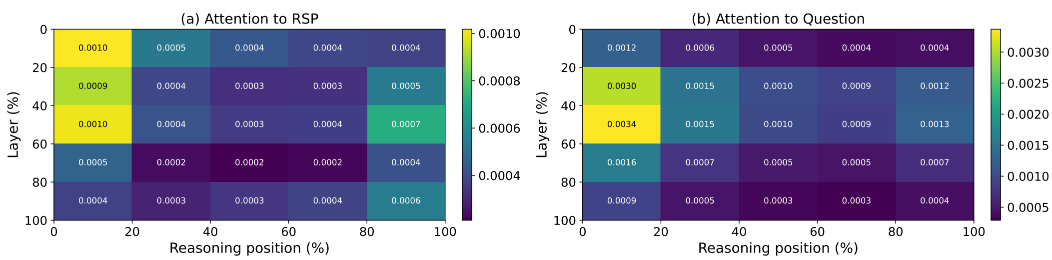
Excluding the final two layers keeps the heatmap focused on reasoning-phase dynamics instead of readout-phase dynamics. This exclusion is not load-bearing for the main claim: the sequence-direction attenuation emphasized in the main text is also visible without it, and Theorem 1 concerns sequence-direction attenuation rather than layer-wise monotonicity.

E.5 Additional Suffix Heatmaps

Figure 5 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across both models: question-token attention varies broadly across layers, whereas RSP attention shows the layer-wise and sequence-wise decay predicted by Theorem 1.



(a) Qwen2.5-Math-1.5B-Instruct, suffix



(b) LLaMA-3.1-8B-Instruct, suffix

Figure 5: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 1: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

F GRPO Formulation

For a given prompt q, the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of G responses $\{o_i\}_{i=1}^{G}$ with corresponding rewards $\{R_i\}_{i=1}^{G}$. The advantage of each response is computed by normalizing the group-level rewards:

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^{G})}{\text{std}(\{R_i\}_{i=1}^{G})}. \tag{9}$$

The policy is updated via a clipped objective with a KL penalty:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big(\min\Big(r_{i,t}\hat{A}_{i,t},\ \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon)\,\hat{A}_{i,t}\Big) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\Big)\right], \tag{10}$$

where $r_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t})/\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ is the importance sampling ratio, ε is the clipping range, and β controls the strength of the KL penalty against the reference policy π_{ref}.

G Broader Impacts

This work provides empirical and theoretical analysis of how random embedding injection affects the reasoning behavior of large language models. The contributions are primarily foundational, with potential downstream implications for reinforcement learning-based post-training of reasoning models such as GRPO.

Positive impacts. RSP-induced trajectory diversity could improve the sample efficiency of RL-based LLM post-training by providing richer learning signals within group-relative advantage estimation. This has the potential to reduce compute costs associated with alignment and reasoning-oriented fine-tuning, making such methods more accessible to research groups with limited resources.

Potential concerns. Methods that expand the effective output support of LLMs may increase the reachability of low-probability tokens, which includes both desirable alternative reasoning paths and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems should combine it with existing safety filtering and alignment mechanisms rather than treating it as a standalone diversity source.

Limitations on impact scope. Since this work focuses on analysis rather than deploying a new model or dataset, the direct societal impact is limited. The broader implications described above are contingent on future work integrating RSP into applied training pipelines.

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